Kimia Hamidieh

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Education

Massachusetts Institute of Technology (MIT) Cambridge, MA
Ph.D. in Computer Science (EECS Department) *Sep. 2023 – Present*
Advisor: Marzyeh Ghassemi

University of Toronto Toronto, ON
M.Sc. in Computer Science *Sep. 2020 – Sep. 2023*
Advisor: Marzyeh Ghassemi

Sharif University of Technology Tehran, Iran
B.Sc. in Computer Engineering *Sep. 2016 – Aug. 2020*

Research Experience

Microsoft Research New England Cambridge, MA
Research Intern *May 2025 – Aug. 2025*
Host: David Alvarez-Melis

- Formalized and quantified data synergy and interference in language model pretraining by proposing domain-aware scaling laws and analyzing scaling behavior under different pretraining data mixtures.

Google DeepMind Mountain View, CA
Student Researcher *May 2024 – Aug. 2024*
Host: Dara Bahri

- Developed active prompt selection methods for online direct alignment to reduce annotation budget while maintaining performance under limited human feedback.

Healthy ML Group (MIT CSAIL/LIDS & IMES) Cambridge, MA
Graduate Research Assistant *Sep. 2022 – Present*
Advisor: Marzyeh Ghassemi

- PhD work on Data-centric approaches to Reliable & Safe AI.

Institute of Science and Technology Austria (ISTA) Klosterneuburg, Austria
Research Intern *Jul. 2019 – Sep. 2019*
Advisor: Christoph Lampert

- Studied functional vs. parametric equivalence classes of models and implications for redundancy in parametrization.

Publications & Preprints

- Kimia Hamidieh, Lester Mackey, David Alvarez-Melis. *"Domain-Aware Scaling Laws Uncover Data Synergy"*. Preprint. Accepted to NeurIPS 2025 workshop on Evaluating the Evolving LLM Lifecycle.
- Kimia Hamidieh, Veronika Thost, Walter Gerych, Mikhail Yurochkin, Marzyeh Ghassemi. *"Complementing Self-Consistency with Cross-Model Disagreement for Uncertainty Quantification"*. Preprint (under review).
- Saachi Jain*, Kimia Hamidieh*, Kristian Georgiev*, Andrew Ilyas, Marzyeh Ghassemi, Aleksander Madry. *"Data Debiasing with Datamodels (D3M): Improving Subgroup Robustness via Data Selection"*. NeurIPS 2024. [* denotes equal contribution]
- Walter Gerych, Haoran Zhang, Kimia Hamidieh, Eileen Pan, Maanas Sharma, Thomas Hartvigsen, Marzyeh Ghassemi. *"Test-Time Debiasing of Vision-Language Embeddings"*. NeurIPS 2024.

- Kimia Hamidieh, Haoran Zhang, Walter Gerych, Thomas Hartvigsen, Marzyeh Ghassemi. *"Identifying Implicit Social Biases in Vision-Language Models"*. AIES 2024. [**Oral Presentation**]
- Kimia Hamidieh, Haoran Zhang, Swami Sankaranarayanan, Marzyeh Ghassemi. *"Views can be deceiving: Improved SSL through feature space augmentation"*. ICLR 2024. [**Spotlight Presentation, top 5% of the submissions**]
- Natalie Dullerud, Karsten Roth, Kimia Hamidieh, Nicolas Papernot, Marzyeh Ghassemi. *"Is Fairness Only Metric Deep? Evaluating and Addressing Subgroup Gaps in Deep Metric Learning"*. ICLR 2022.
- Stephan Rabanser, Anvith Thudi, Kimia Hamidieh, Adam Dziedzic, Nicolas Papernot. *"Selective Classification Via Neural Network Training Dynamics"*. Preprint.
- Aparna Balagopalan, Haoran Zhang, Kimia Hamidieh, Thomas Hartvigsen, Frank Rudzicz, Marzyeh Ghassemi. *"The Road to Explainability is Paved with Bias: Measuring the Fairness of Explanations"*. ACM FAccT 2022.
- Amin Ghareyazi, Amirreza Kazemi, Kimia Hamidieh, Hamed Dashti, Maedeh Sadat Tahaei, Hamid R. Rabiee, Hamid Alinejad-Rokny, Abdollah Dehzangi. *"Pan-cancer integrative analysis of whole-genome De novo somatic point mutations reveals 17 cancer types"*. BMC Bioinformatics.

Honors & Awards

- Finalist for Kempner Institute Research Fellowship (Harvard University), 2025.
- Vector Institute Research Grant, 2021–2022.
- ISTA Research Intern Fellowship, Summer 2019.
- Member of Iranian National Elite Foundation, Sep. 2016 – Aug. 2020.
- Top 0.1% in Iranian National University Entrance Exam, Jun. 2016.
- Member of National Organization for Development of Exceptional Talents, 2009 – 2016.

Teaching & Service

Teaching Assistant
AI for Society (6.3950, MIT) — *Fall 2025*

Teaching Assistant
Machine Learning (6.790, MIT) — *Fall 2023*

Teaching Assistant
Introduction to Machine Learning (CSC311, UofT) — *Fall 2021*

Teaching Assistant
Introduction to AI (CSC384, UofT) — *Spring 2021*

Reviewer (ML Conferences)
NeurIPS 2023, 2024, 2025; ICLR 2024, 2025, 2026; ICML 2025; AISTATS 2025, 2026

Reviewer (AI Ethics & ML for Health)
FAccT 2023; CHIL 2023, 2025; ML4H 2024

Skills

- **Languages:** Python, C/C++, Java, R, Bash.
- **Frameworks & Tools:** PyTorch, JAX, Slurm, Azure, Git, Docker.